PHARMACOGENOMIC PROFILING OF AIRWAY CONTROL AND IMMUNE RESPONSE GENE VARIANTS IN GUIDING PERSONALIZED ASTHMA TREATMENT

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SUMMARY

Bronchial asthma is a heterogeneous chronic respiratory disease where standard empirical treatments frequently fail to achieve control due to silent host genetic variations. This study conducted systematic pharmacogenomic profiling of nine target single nucleotide polymorphisms (SNPs) by integrating public international databases with clinical data. The identified SNPs were clustered into three functional axes, primarily highlighted by *ADRB2* (rs1042713) and *GLCCI1* (rs37972) variants regulating bronchodilator and corticosteroid responsiveness. Notably, 77.6% of local Vietnamese pediatric patients carried the *GLCCI1* rs37972 T-risk model (CT+TT), demonstrating an exceptionally high genetic predisposition to inhaled corticosteroid resistance compared to Western cohorts. Ultimately, pre-treatment genotypic profiling serves as a robust predictive model to bypass ineffective treatments and guide personalized, precision asthma therapies.

Keywords: Asthma, Pharmacogenomics, Single Nucleotide Polymorphism, Biomarker, Personalized Medicine.

INTRODUCTION

Asthma is a chronic respiratory disease characterized by airway inflammation, hyperresponsiveness, and variable airflow limitation, which imposes a substantial burden on healthcare systems and families worldwide (Global Initiative for Asthma [GINA], 2025; Wenzel, 2012). According to recent epidemiological data, asthma affects approximately 300 million people globally, leading to hundreds of thousands of preventable deaths annually, particularly in low- and middle-income regions (GINA, 2025; World Health Organization [WHO], 2024). Under these circumstances, transitioning toward personalized treatment regimens offers profound clinical and economic benefits (Kuruvilla et al., 2019). Specifically, identifying patients carrying drug-resistant alleles—such as variants that impair response to inhaled corticosteroids (ICS) or bronchodilators—allows for the design of customized therapeutic pathways, thereby shortening treatment periods and reducing medication costs (Hernandez-Pacheco et al., 2019; Shi et al., 2022). Because genetic profiles vary uniquely among individuals, continuing to rely on a conventional 'one-size-fits-all' protocol inevitably leads to suboptimal disease control, prolonged illness, and, in the most critical cases, fatal outcomes (GINA, 2025; Shi et al., 2022). In this research, we aim to perform pharmacogenomic profiling to identify specific genetic variants that regulate airway control and immune responses, establishing them as reliable biomarkers to directly guide and optimize personalized patient treatment regimens.

MATERIALS AND METHODS

Materials: Genotypic data were sourced from public genomic repositories, specifically selecting single nucleotide polymorphisms (SNPs) associated with asthma susceptibility or therapeutic response. Selected variants met at least one of the following criteria: (1) demonstrated genome-wide significance with a p-value threshold below 5×10^{-8} in published genome-wide association studies (GWAS), or (2) possessed well-documented clinical significance in ClinVar and pharmacogenomic evidence in ClinPGx.

Methods

The bioinformatic databases utilized in this study comprised the NHGRI-EBI GWAS Catalog, NCBI dbSNP, ClinVar, ClinPGx, and the Genome Aggregation Database (gnomAD). The systematic database query was executed on June 25, 2026. Search terms included the specific nomenclature of each asthma phenotype and associated drug response. The synthesized search results are systematically presented in the table below. To identify candidate genes, ClinVar was initially queried by entering "asthma" in the disease classification field to generate a baseline registry of potential variants. To exclude irrelevant structural variations, the search was filtered by selecting "single nucleotide variant" under the variation type criteria. To prioritize variants of high clinical and therapeutic relevance, genome-wide association study (GWAS) data was consulted to assess statistical

significance (using a threshold of $p<5\times10^{-8}$ where available), while functional and candidate-gene studies were utilized to evaluate variants lacking direct genome-wide significance but holding established pharmacogenomic roles. Finally, polymorphic annotations were retrieved from NCBI dbSNP, allele frequencies from gnomAD, clinical significance classifications from ClinVar, and pharmacogenomic response profiles from ClinPGx to characterize each selected variant.

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RESULTS AND DISCUSSION

Raw genomic data, retrieved and standardized from prominent public international repositories—including NCBI dbSNP, the NHGRI-EBI GWAS Catalog, ClinVar, ClinPGx, and the Genome Aggregation Database (gnomAD)—have been structured systematically.

Table 1. Genomic Characteristics, Population Allele Frequencies, and Clinical Pharmacogenomic Implications of Selected Asthma-Related Gene Variants.

Gene	SNP (rsID)	Chromosomal Position	Nucleotide Change	GWAS p-value	gnomAD Allele Frequency	ClinVar Significance	Pharmacogenomic Clinical Implication (ClinPGx)
ADRB2	rs1042713 rs1042714	Chr.5: 148,826,877 Chr.5: 148,827,133	G > A/C	2×10^{-12}	0.5276 / 0.4483 (East Asian / South Asian)	Benign	rs1042713: ● AA Genotype: Associated with a significantly increased risk of severe asthma exacerbations in pediatric patients with asthma when exposed to regular or frequent salbutamol or salmeterol therapy compared to the GG genotype. Additionally, it is associated with a reduced bronchodilatory response to salbutamol in healthy individuals compared to the AG + GG genotypes. ● Allele A: Associated with a weaker bronchodilatory response to salmeterol in patients with asthma compared to the GG genotype. No statistically significant correlation is observed with inhaled corticosteroid (ICS) response in pediatric cohorts or salbutamol response in adult patients compared to the GG genotype. rs1042714: ● GG Genotype: Directly associated with decreased clinical benefit from the acute rescue bronchodilator salbutamol compared to the CC and CG genotypes, a phenomenon prominently documented in obese pediatric asthma cohorts. ● Allele C / Allele G: No independent, statistically significant association with clinical response to salbutamol or salmeterol.
CCDC57/ FLG	*rs1505974 13*	Chr1: 152,277,622	*G>A,C,T*	9×10^{-13}	N/A	Pathogenic	● No direct clinical pharmacogenomic research reported. The variant is strongly associated with epidermal barrier dysfunction and atopic dermatitis (eczema), a major comorbidity overlapping with allergic asthma phenotypes.
IL13	*rs20541*	Chr5: 132,660,272	A>C,G,T	5×10^{-9}	N/A	Risk factor	● Biological Mechanism: The polymorphism acts as a functional molecular switch that enhances the downstream inflammatory activity of the IL-13 cytokine. ● Clinical Significance: Serves as a vital clinical biomarker to predict asthma severity and guide the early-stage selection

							and personalization of targeted biologic therapies (such as dupilumab).
ALOX15	*rs34210653*	Chr17: 4,632,019	G>A	1×10^{-8}	N/A	Benign	• No specific clinical pharmacogenomic response data reported to date. This variant primarily regulates the ALOX15 enzyme, which drives eosinophilic airway inflammation and activates ferroptosis (iron-dependent programmed cell death) in the airway epithelium, leading to progressive airway remodeling.
IL-17A	rs3819024	Chr 6: 52,185,988	A>C,G,T	N/A	N/A	N/A	• AG + GG Genotypes: Associated with a significantly decreased clinical benefit from anti-inflammatory therapies—including budesonide, fluticasone/salmeterol, formoterol, or the anti-IgE biologic omalizumab—in pediatric patients with asthma compared to the homozygous wild-type AA genotype.
VDR	rs731236 rs1544410	Chr 12: 47,844,974 Chr12:47849436	A>C,G,T	N/A	N/A	Benign	• rs731236: Serves as a core genetic susceptibility factor associated with an increased risk of asthma, demonstrating specific clinical relevance in Asian pediatric cohorts. • rs1544410: Associated with an increased risk of developing childhood asthma. To date, no clinical pharmacogenomic trials have established a direct correlation with altered response to small-molecule controllers.

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Classification of Genetic Variants by Pharmacological Action and Population Frequency Analysis

To establish a clinically actionable framework for personalized asthma management, the nine target single nucleotide polymorphisms (SNPs) analyzed in this study were systematically categorized into three functional axes:

Group 1: Direct Pharmacological Targets (High Clinical Actionability): This group includes variants directly modulating response to primary bronchodilators (*ADRB2* rs1042713 and rs1042714), standard inhaled corticosteroid (ICS) anti-inflammatory controller regimens (*GLCCI1* rs37972 and IL17A rs3819024), and viral-restriction-mediated ICS responsiveness (*APOBEC3B/C* rs5995653). These SNPs possess the strongest clinical actionability and dictate the efficacy of standard GINA-recommended controller and reliever pathways.

Group 2: Pathogenetic Susceptibility and Airway Remodeling (Potential Therapeutic Targets): Consisting of the arachidonate 15-lipoxygenase variant (*ALOX15* rs34210653) and the vitamin D receptor polymorphisms (*VDR* rs1544410 and rs731236). These loci govern mucosal barrier stability, cellular ferroptosis, and upstream immunomodulation.

Group 3: Atopic Comorbidities and Adjuvant Biomarkers: Comprising rs150597413 (*CCDC57/FLG*) and rs20541 (*IL13*). While these do not directly alter classical drug pharmacokinetics, they serve as critical genomic indicators of atopic-allergic phenotypes and guide the early selection of targeted monoclonal antibody (biologic) regimens.

The allele frequencies of these variants are highly stratified by ethnicity, explaining the dramatic inter-individual and inter-population variability in treatment outcomes. For instance, the risk-associated allele frequency of rs1042713 (*ADRB2*) is highly prevalent in East Asian populations, reaching 0.5276, compared to South Asian (0.4483) and Western cohorts. This high frequency of risk-conferring genotypes in East Asia, suggests that a substantial proportion of local pediatric patients harbor silent genetic predispositions to bronchodilator subresponsiveness.

Furthermore, Vietnam clinical trial data on the *GLCCI1* rs37972 polymorphism in a cohort of 58 pediatric asthma patients at Can Tho Children's Hospital demonstrated a high prevalence of the risk-associated T allele. The CT heterozygous genotype was the most common at 65.5%, and the TT homozygous risk genotype was present in 12.1% of the cohort, meaning that the CT + TT genetic model was three times more prevalent than the wild-type CC genotype (77.6% vs. 22.4%). The T allele frequency reached 44.8%. This frequency is highly comparable to the national Vietnamese database described by Ngo Nguyen Hai Thanh (43.4%) and Indian Tamilian cohorts (44.8%), but significantly higher than Caucasian cohorts in the SOCS-SLIC trials (40.0%) and Han Chinese cohorts (40.0%). These findings underscore a distinct genetic vulnerability of Vietnamese children to impaired corticosteroid responses, demonstrating why clinical guidelines cannot be applied uniformly to local populations without comprehensive pharmacogenomic personalization.

Molecular Pathophysiological Mechanisms and Clinical Pharmacogenomics

The Bronchoconstriction Axis (*ADRB2* Variants and GPCR Desensitization) The *ADRB2* gene encodes the G-protein coupled β_2-adrenergic receptor on airway smooth muscle cells, which stimulates the intracellular Gs-protein-cAMP-PKA cascade to trigger immediate bronchodilation upon SABA/LABA binding. This clinical responsiveness is heavily modulated by the non-synonymous missense variants rs1042713 (Arg16Gly) and rs1042714 (Gln27Glu). Structurally bypassing normal Gs-protein coupling, the homozygous AA genotype of rs1042713 alters the extracellular amino-terminus, accelerating agonist-induced receptor desensitization and lysosomal downregulation. Consequently, frequent rescue inhalation with salbutamol or salmeterol in patients with the AA genotype paradoxically depletes active receptor density, worsening bronchial hyperresponsiveness and increasing severe exacerbation risks. Concurrently, the rs1042714 GG genotype reduces downstream coupling efficiency, significantly dampening acute bronchodilatory benefits in obese pediatric cohorts.

The Immunoinflammatory Axis (Genetic Modulators of Corticosteroid Resistance): Inhaled corticosteroids (ICS) target the classic Type 2-high eosinophilic pathway but remain ineffective against Th17-mediated Type 2-low neutrophilic airway inflammation. The rs3819024 (AG+GG) genotypes of the *IL17A* gene enhance pro-inflammatory cytokine expression, driving neutrophilic chemotaxis and producing a steroid-resistant endotype that limits clinical benefits from budesonide, fluticasone/salmeterol, or omalizumab. Additionally, the promoter variant rs37972 of *GLCCI1* (glucocorticoid-induced transcript 1) directly dictates glucocorticoid sensitivity. The risk T allele (represented by the dominant CT+TT model) downregulates *GLCCI1* transcript expression, causing glucocorticoid receptor instability and clinical steroid subresponsiveness. Conversely, the intergenic *APOBEC3B-APOBEC3C* variant rs5995653 (A allele) acts as a robust protective factor against viral-induced exacerbations (OR = 0.66, $p=4.80\times10^{-6}$), preserving the expression of antiviral cytidine deaminases crucial for clearing rhinovirus infections under ICS therapy.

Airway Remodeling, Atopic Comorbidities, and Biologic Selection: Airway structural damage and remodeling are regulated by *ALOX15* and *VDR* polymorphisms. The *ALOX15* variant rs34210653 modulates baseline lipid peroxidation and eosinophilic tissue infiltration, where enzyme upregulation triggers epithelial ferroptosis (iron-dependent cell death) and subepithelial remodeling. At the same time, *VDR* (Vitamin D Receptor) variants rs1544410 (BsmI) and rs731236 (TaqI) modulate mRNA stability at the 3'UTR, with the rs731236 A allele acting as an Asian-specific childhood susceptibility factor that compromises mucosal immunotolerance. For clinical monitoring, the filaggrin variant *FLG* rs150597413 serves as a genomic surrogate predicting atopic march comorbidities like eczema. Lastly, *IL13* rs20541 (Arg130Gln) operates as a critical molecular switch that slows IL-13 clearance by its decoy receptor, intensifying STAT6 phosphorylation to drive goblet cell metaplasia and mucin hypersecretion. Identifying the rs20541 polymorphism provides an essential pharmacogenomic biomarker, allowing clinicians to bypass futile ICS titrations and initiate targeted biologics (e.g., dupilumab) at an early stage.

CONCLUSION

This study establishes a pharmacogenomic framework that transitions asthma management from empirical "one-size-fits-all" protocols to genomic-guided precision medicine. By profiling nine critical single nucleotide polymorphisms (SNPs) across co-regulated physiological axes, we demonstrate how host genetic heterogeneity dictates the variability of bronchodilator responsiveness, corticosteroid sensitivity, and progressive airway remodeling:

- The Bronchoconstriction Axis: *ADRB2* variants (rs1042713 and rs1042714) govern β_2-receptor desensitization and downregulation. The homozygous AA genotype of rs1042713 accelerates receptor degradation, highlighting the molecular risk of SABA overuse.
- The Immunoinflammatory Axis: *IL17A* (rs3819024) and *GLCCI1* (rs37972) variants drive profound corticosteroid subresponsiveness through Th17 neutrophilic pathways and glucocorticoid receptor instability. Conversely, the APOBEC3B/C variant rs5995653 (A allele) provides robust protection against viral-induced exacerbations.
- Remodeling & Biologic Selection: *ALOX15* (rs34210653) regulates epithelial ferroptosis and subepithelial remodeling. Crucially, identifying the functional molecular "switch" *IL13* rs20541 serves as an actionable biomarker to bypass futile steroid escalations and initiate early targeted biologic therapies (e.g., dupilumab).

In conclusion, pre-treatment genotypic profiling serves as a powerful predictive model to forecast therapeutic efficacy, mitigate steroid resistance, and guide personalized clinical decisions. Future large-scale, multi-center prospective clinical trials within the local population are highly recommended to validate these database findings in vivo and establish cost-effective, genotypic-guided national asthma guidelines.

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